

COURSE PROJECT

Research Replication & Extension

Advanced Natural Language Processing

2–3

Team Size

Week 12

Phase 1

Finals

Phase 2

2022

Papers: After

Project Overview

This two-phase project challenges teams to engage deeply with current NLP research, first by understanding and communicating a published contribution, then by replicating and extending it. The goal is to develop the full research cycle: reading critically, understanding methodology, implementing from scratch, and contributing original improvement.

Paper Selection Rule

All papers must be published after 2022. Verify the publication year on ACL Anthology, arXiv, or the conference proceedings before submitting your selection. Papers from ACL, EMNLP, NAACL, NeurIPS, ICML, or ICLR are acceptable.

Phase 1 — Choosing your Paper

Objective

Each team selects one NLP paper from a recognized venue (see approved sources), thoroughly understands it.

Paper Selection

Papers must be selected from the following sources:

- ACL Anthology venues: ACL, EMNLP, NAACL, COLING, CoNLL, EACL, TACL
- Top ML/AI venues with NLP content: NeurIPS, ICML, ICLR, AAAI
- arXiv preprints are acceptable only if the work was subsequently accepted at one of the above

No Duplicate Papers

Each paper may only be claimed by one team. Paper selection is first-come, first-served via the course portal. Once a paper is claimed, it will be marked unavailable. Teams are encouraged to have 2–3 backup options ready.

Phase 2 — Replication & Extension

Objective

Teams implement the core model from their Phase 1 paper, verify that their implementation reproduces the reported results (or comes reasonably close given resource constraints), **and then propose at least one meaningful extension or improvement (as future work, no need to implement)**. The final deliverable is a written paper in EMNLP format and a clean, well-documented GitHub repository, along short demo video.

What Counts as a Meaningful Extension

Extensions should go beyond hyperparameter tuning. Acceptable extension categories include:

- Architectural modification: For example, **replacing** an attention mechanism, modifying the encoder, adding a new module.
- Task transfer: **Applying** the model to as different language, domain, or downstream task not explored in the original.
- Data augmentation or efficiency: **Improving** sample efficiency, reducing compute, or applying the method to a low-resource setting.
- **Combining ideas**, integrating a technique from a second paper to complement the original model.
- Robustness analysis, systematically evaluating failure modes and proposing a targeted fix.

Scope Calibration

You are not expected to match or exceed state-of-the-art results. A thoughtful, well-reasoned extension with honest analysis of what worked and what did not is far more valuable than chasing benchmark numbers. Negative results written up rigorously are fully acceptable.

HAVE FUN with THIS PROJECT. That is what matters to me.

Deliverable 1 — EMNLP Format Paper

The paper should be written as if submitted to EMNLP. Use the official ACL/EMNLP LaTeX or Word template (available at <https://2024.emnlp.org/calls/papers/>).

Section	Expectations
Abstract	150–200 words. State the original paper, your replication result, and the extension plan for future work.
Introduction	Motivate the original work and clearly state what your team contributes.
Related Work	Situate the paper in the literature; cite at least 8 relevant works .
Background	Explain the model architecture and training setup of the original paper.

Section	Expectations
Replication	Describe your implementation decisions. Report results vs. original, with any gaps explained.
Extension and Future Work	Describe your proposed improvement, rationale, implementation plans. This section is the plan for future work. There is 20 bonus points if you implement your extension idea.
Conclusion	Summarize findings and suggest future directions.

Page limit: **No page limit** + unlimited references. Appendices are allowed but will not be read by default.

Deliverable 2 — GitHub Repository

The repository must be made public by the final submission deadline and include:

- README.md with: project title, team members, paper reference, setup instructions, and a results table
- requirements.txt or environment.yml for reproducibility
- Modular, commented code, scripts for training, evaluation, and running the extension
- A results/ or outputs/ folder with logs or sample outputs from at least one full run
- Any model checkpoints or instructions for downloading them (e.g., via HuggingFace Hub)

Code Quality Note

Reviewers will attempt to run your code. Repositories with no README, broken imports, or uncommented code will lose points under the Code Quality rubric regardless of research quality. Treat it as a public portfolio piece.

Submission — Phase 2

- **EMNLP-format paper as PDF, submitted via Canvas**
- **GitHub repository link (must be public at submission time)**
- **Short demo video (5–10 min) walking through a key result or the extension, team members should split narration**

Grading Rubric

Phase 2 — Replication & Extension

Criterion	Points	Description
Replication Fidelity	15	Implementation reproduces key results; gaps are documented and explained
Extension Plan Quality	20	Extension is well-motivated, non-trivial, and results are clearly reported
Paper Writing	15	Well-structured EMNLP-format paper; clear prose; proper citations
Analysis Depth	10	Error analysis, ablations, or qualitative examples beyond aggregate scores
Code Quality & Repo	10	Public repo; runs with provided instructions; clean and documented

Total: 100 points (weighted by phase percentages above)

Academic Integrity & Use of AI Tools

You are encouraged to use LLM-based tools (e.g., GitHub Copilot, ChatGPT) for coding assistance and writing support. However:

- All experimental results must be produced by your own code and runs, you may not report results from any other source
- The paper must be written by your team. AI-assisted drafts must be substantially revised; directly submitting AI-generated text is not acceptable
- Any use of AI writing tools must be disclosed in an Author Contributions statement at the end of the paper
- Code copied from external repositories must be cited in comments and in the README

Questions?

Post on the course discussion board or bring them to office hours.